



UNIVERSITY OF BAHRAIN
COLLEGE OF INFORMATION TECHNOLOGY
CYBERSECURITY SENIOR PROJECTS



Thaghras : Practical Cybersecurity Learning Platform Through Packet Analysis

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ABSTRACT

Beginner cybersecurity learners frequently face difficulties when attempting to translate theoretical networking knowledge into practical packet-analysis and traffic-investigation skills. Many educational platforms offer only limited exposure to realistic network traffic and practical cybersecurity tools, which can negatively affect learner confidence and reduce hands-on understanding. This research introduces Thaghras, a web-based cybersecurity learning platform designed to support packet analysis, protocol investigation, and Wireshark-based training through downloadable PCAP challenges and AI-generated investigation scenarios.

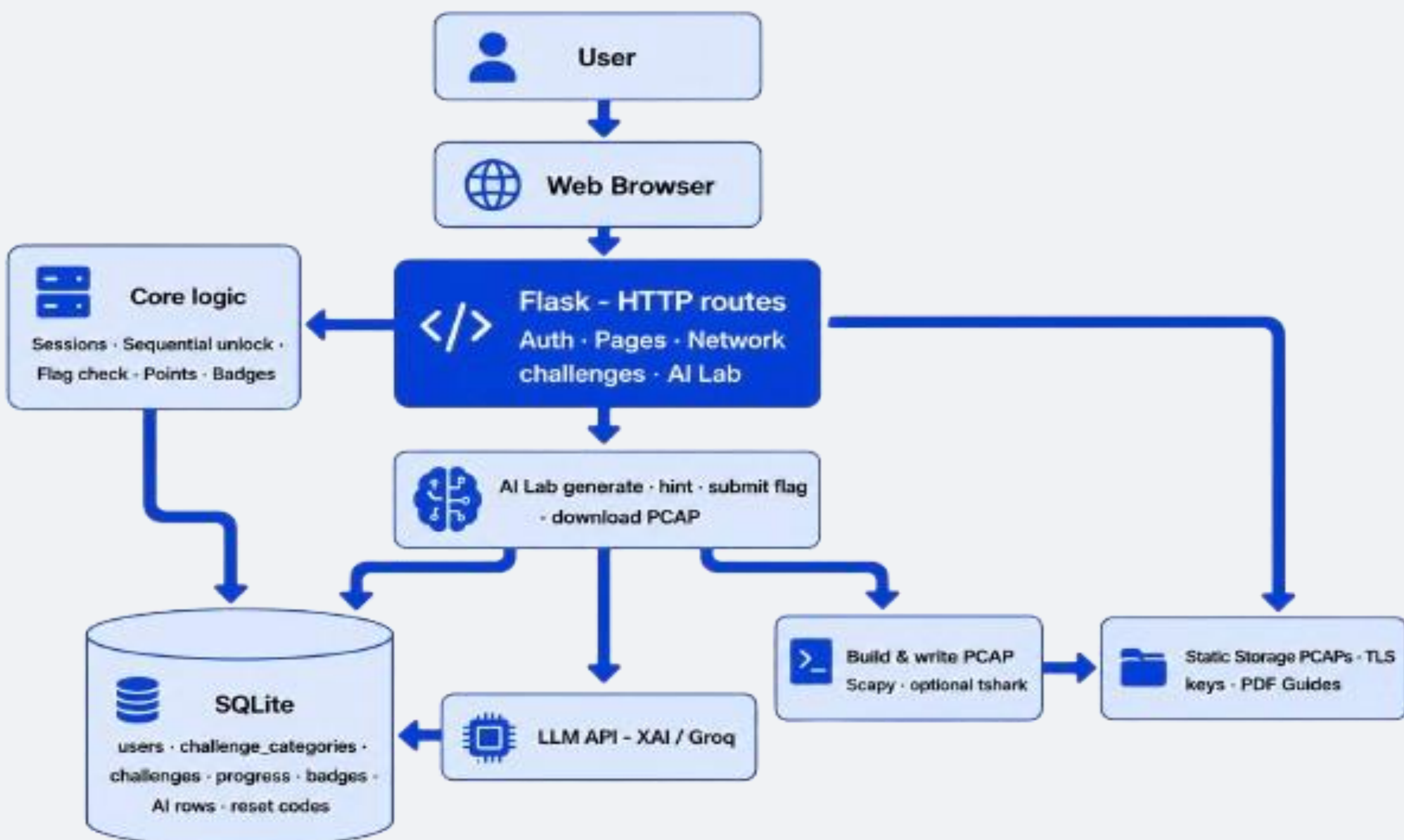
The platform was implemented using Flask, SQLite, Scapy, and configurable AI-provider integration. Thaghras delivers a structured curriculum consisting of 40 packet-analysis challenges distributed across multiple protocol categories, in addition to an AI Lab capable of creating dynamic packet-analysis scenarios from natural-language prompts. The system further incorporates sequential challenge unlocking, flag-validation mechanisms, scoring and badge systems, integrated protocol guides, and downloadable PCAP-based investigation workflows.

The findings demonstrate that integrating structured challenge progression, practical investigation activities, and gamified learning features can strengthen learner engagement and enhance practical cybersecurity education. The project also demonstrates that AI-generated investigation scenarios can increase training diversity while maintaining realistic packet-analysis workflows within a controlled educational environment.

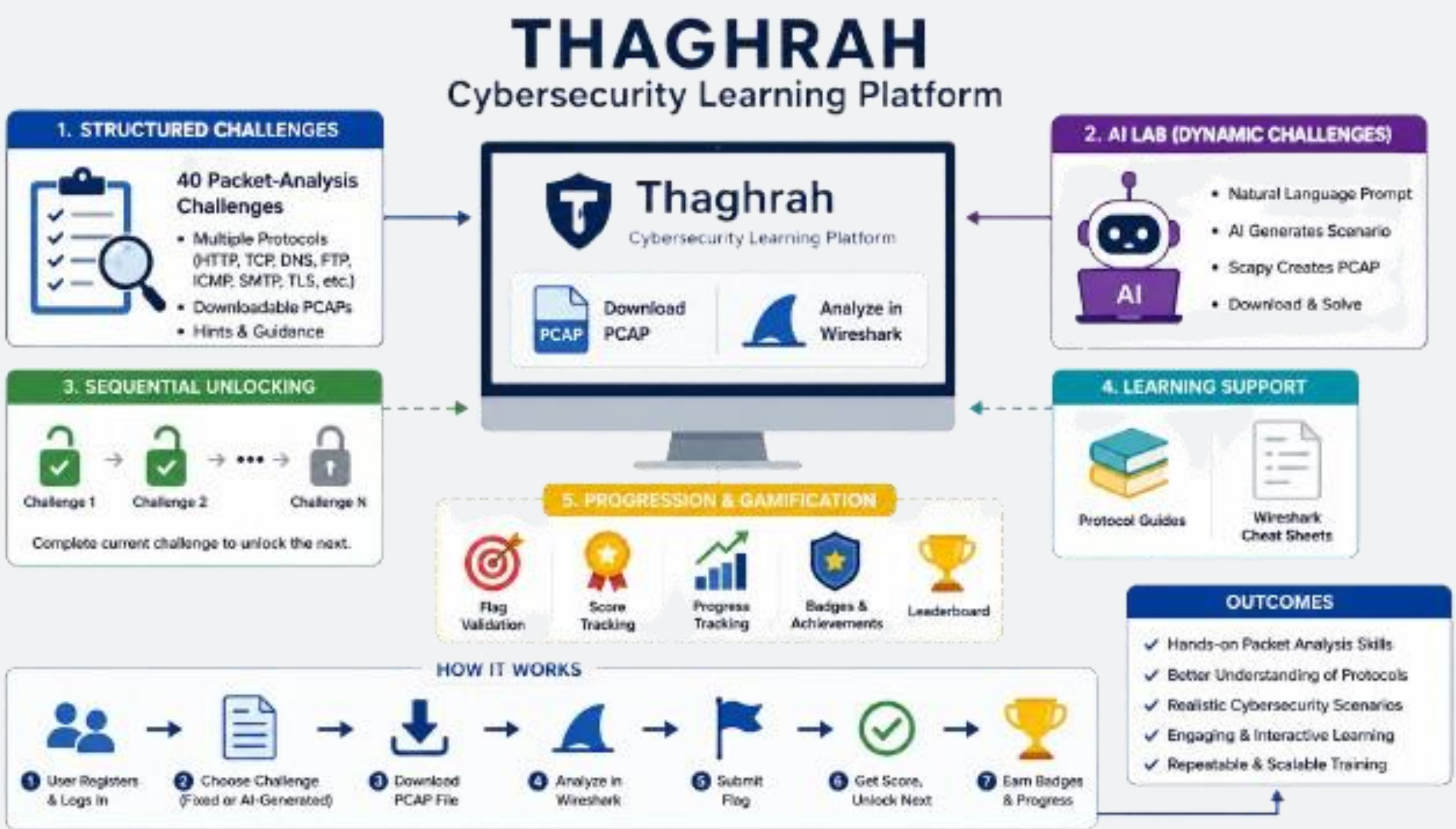
OBJECTIVES

- Help beginner cybersecurity learners develop practical packet-analysis skills.
- Bridge the gap between theoretical networking knowledge and hands-on investigation.
- Provide realistic PCAP-based cybersecurity challenges using protocol-focused exercises.
- Encourage manual analysis and critical thinking instead of automated solving.
- Support learner progression through structured challenges and progress tracking.
- Integrate AI-generated scenarios to increase challenge variety and training realism.

SYSTEM ARCHITECTURE



RESULTS



CONCLUSION & FUTURE WORK

- Conclusion**
- The evaluation confirmed that the implemented workflows operated successfully within the intended educational environment.
 - The platform correctly supported challenge progression, flag validation, downloadable PCAP functionality, security protections, performance behavior, and AI-generated challenge creation.
 - The system effectively provided structured packet-analysis learning through guided forensic investigation exercises.
- Future Work**
- Expand challenge categories beyond network-analysis challenges.
 - Improve the consistency and quality of AI-generated challenges.
 - Support additional cybersecurity concepts beyond encoding, encryption, fragmentation, and brute-force scenarios.
 - Introduce real-time packet-capture analysis alongside downloadable PCAP workflows.